

Curriculum Vitae

| Experience

Machine Learning Engineer

Bharti Airtel Limited, Gurugram

One of India's largest telecom providers serving 650M+ customers globally across 17 countries with 21.56B USD revenue and 24,420 employees.

August 2025 – Present

- Designed and built a real-time conversational voice bot that helps field engineers resolve broadband complaints at customer premises — architecting a Meta OmniASR (STT), Qwen3.5 (LLM), and Voxtral (TTS) turn-taking loop in LangGraph across decision, agent, and communication nodes, achieving sub-1,000 ms round-trip latency. Built the asynchronous FastAPI microservices and a MongoDB-backed priority call queue and state machine to manage concurrent live calls over Airtel IQ WebSocket telephony.
- Finetuned Meta's Omnilingual ASR (OmniASR 1B/3B) and Whisper-Apex for Hindi/Hinglish speech, engineering a μ -law + noise augmentation pipeline that injected real noise profiles mined from in-house customer call recordings into clean training data to match production telephony conditions; reached ~18.9% WER / ~9.0% UER on a held-out validation set.
- Engineered the ASR serving infrastructure across two lanes — OmniASR (wav2vec2 + LLM-prefix) on a dedicated NVIDIA Triton lane and Whisper-Apex on vLLM — with a real-time audio front-end using Silero VAD for speech segmentation and DeepFilterNet noise suppression, plus bandwidth-extension stages to close a confirmed 8 → 16 kHz telephony audio ceiling.
- Revamped the WCL stack (2G/4G/5G) and built the team's first Docker-based Apache Airflow, CI/CD-enabled orchestration — migrating pipelines between PySpark and pandas by workload profile and orchestrating 20+ ETL workflows as SME, cutting runtime 75% and compute cost 85%.
- Leading nationwide Duct operations across 12+ states — a business-critical program directly affecting customer network performance, traffic, and tower/cell operations — as tech lead and primary SME/POC coordinating cross-functional teams India-wide; deployed a real-time monitoring dashboard that eliminated manual inspections and gave stakeholders instant operational visibility.
- Built large-scale PySpark pipelines processing terabytes of broadband customer data daily across India, generating network-health KPIs, churn signals, and CX metrics for Airtel's flagship Broadband Experience initiative.

Associate Analytics & Model Analyst

Moody's Insurance Solutions, Noida

Provides catastrophe models and risk intelligence for insurers, with 7.09B USD revenue and a workforce of 15,838.

April 2023 – August 2025

- Architected an end-to-end data orchestration framework on Dagster, integrating high-frequency weather data from 30+ global sources with CI/CD automation via Jenkins and GitHub Actions, improving pipeline efficiency by 90%.
- Implemented a Django-based data visualization dashboard using Mapbox, JavaScript, MongoDB, and Python, enabling global wind station tracking and cross-referencing with the Postgres database to identify over 18,000 missing stations.
- Collaborated with the Event Response team to automate Support Center uploads via an API, reducing manual upload time from 40-60 minutes to zero, enhancing efficiency for natural disaster reports and forecasts.
- Built an HWind statistics dashboard using JavaScript, Python, and Django, tracking daily and weekly data feed counts with interactive charts to identify problematic transactions and feed downtimes.

Data Associate

Verisk Analytics (Wood Mackenzie), Gurugram

Global insight business for renewable energy, & natural resources, with 1.9B USD revenue and 2000+ employees.

November 2021 – April 2023

- Enhanced data collection and processing pipelines in Python on AWS SageMaker, improving efficiency by 60%.
- Built Python automation scripts to extract oil price data from country reports, update Excel deliverables, and trigger email notifications on completion, reducing end-to-end processing time by 90%.
- Mentored 5 new team members on codebases, workflows, and tooling; led cross-team project handovers managing roadmaps and task allocation via JIRA.
- Built a Power BI dashboard tracking project delivery metrics across the team, improving visibility into bottlenecks and task completion rates.

Education

B.Tech. in Biotechnology

Amity University, Noida

August 2017 – March 2021

CGPA: 8.15/10

Skills

Languages: Python, SQL, JavaScript

Big Data: Apache PySpark, Pandas, Hadoop, HDFS, YARN, Delta Lake

Orchestration & DevOps: Apache Airflow, Docker, Docker Compose, Dagster, Jenkins, GitHub Actions, Bitbucket

ML/AI: PyTorch, Transformers, HuggingFace, LangGraph, LangChain, LLMs, Speech-to-Text/Text-to-Speech (STT/TTS), ASR finetuning, OmniASR, Whisper, DeepFilterNet, Silero VAD, noise suppression

Model Serving: Triton Inference Server, vLLM

Backend & Web: FastAPI, Django, Streamlit, WebSockets

Databases: PostgreSQL, MongoDB

BI & Visualization: Power BI, Tableau

Tools: Git, Linux, Paramiko, AWS SageMaker

Spoken: English (Proficient), Hindi (Native)

Publications

Identification of vaccine targets & design of vaccine against SARS-CoV-2 coronavirus using computational and deep learning-based approaches

PeerJ · May 2022

Identification of vaccine targets in pathogens and design of a vaccine using computational approaches

Nature Scientific Reports · September 2021

To Study the Effect of Unconventional Treatment Protocol on COVID-19 Patients in Delhi Using Artificial Intelligence Based Methods

SSRN · September 2020

Extracurricular

Member & AVP Education

Toastmasters Club, West Delhi

April 2023 – March 2025

Organized weekly meetings, speaker selection, and onboarded new members.

Volunteer & Drive Lead

Robin Hood Army, Delhi

October 2022 – Present

Led 30+ food distribution drives and taught 25+ students weekly.

Coordinator

CRC Placement Cell, Amity University

February 2019 – March 2021

Orchestrated 100+ on-campus placement drives and pre-placement prep sessions.